Lesson plan · for the teacher

Blanks, zeros and repeats

Data engineering · Level 2 · Lesson 3

Time
40 min

Materials
A paper rain table with four hidden problems (one per group); A cleaning-log sheet: what, where, what we did, why; Optional: a computer or phone with Python

Learning goals

- Tell a blank from a zero, and never fill one silently.
- Find repeats, after the names agree.
- Check every batch, and keep a cleaning log.

Activity: Data detective

1. Warm-up (5 min): Show a week of rain with one 999 in it, and its average. Ask: did it really rain that much? What happened?
2. Teach (8 min): Explain a blank versus a zero, true repeats, and the four checks. Show the three choices when a check fails.
3. Hunt (15 min): Groups search the rain table for the hidden problems and fill in the cleaning log for each one.
4. Recalculate (7 min): Groups work out each station's average with and without the fixes and write both on the board.
5. Wrap-up (5 min): Ask which mistake no check would catch, such as 12 typed as 21. Then give the exit ticket.

Hide the four problems in different stations and days. Keep a copy of the answers, and accept any decision a group can explain and has written in its log.

Discussion

- Why is a blank mark not the same as a zero?
- What did stopping the whole batch cost us?
- What should a cleaning log say?

Exit ticket

- Write one blank, one zero and one repeat you might find in a register, and what you'd do with each.

Try it at school tomorrow
Play "Data detective": give groups a paper rain table with four hidden problems (a blank, a 999, a minus sign and a repeat). They find them, decide what to do, and write a cleaning log.

Practice questions · for you

Blanks, zeros and repeats

Data engineering · Level 2 · Lesson 3

1 **A batch has one bad reading among twenty good ones. Which choice keeps the good data moving?**

◯ Set the bad row aside to check ◯ Let everything through ◯ Stop the whole batch

2 **A blank rain reading means it didn't rain that day.**

◯ True ◯ False

3 **marks.csv has two blank maths marks, and Musa really scored 0. Change the program so it counts the blanks and leaves them out of the average: it should print 2, then 10.75.**

```
import csv
blanks = total = count = 0
with open("marks.csv") as f:
    for row in csv.DictReader(f):
        mark = row["maths"]
        total = total + int(mark or 0)
        count = count + 1
print(blanks)
print(total / count)
```

Write your program:

4 **Match each check to a fault it catches.**

In range (0 to 300 mm)	• • • •	Tuesday's reading sent twice
Not empty	• • • •	"Hiltop" instead of Hilltop
No repeats	• • • •	999, the gauge's error code
Known name	• • • •	A reading nobody wrote down

5 **rain.csv has a reading sent twice for two stations. Finish the program so it prints each repeat (station and day) once, as it finds it: it should print Hilltop Tue, then Market Wed.**

```
import csv
seen = set()
with open("rain.csv") as f:
    for row in csv.DictReader(f):
        key = (row["station"], row["day"])
        if key in seen:
            print(row["station"], row["day"])
```

Write your program:

✎ **Think of a report your school sends. Where could blanks or repeats hide in it, and what would you write in a cleaning log?**

Check your answers

Blanks, zeros and repeats

Data engineering · Level 2 · Lesson 3

1 **Set the bad row aside to check**

Setting the bad row aside lets the twenty good readings reach the report, while the bad one waits for a person to check it with the club.

2 **False**

A blank means nobody read the gauge; zero means it was read and was dry.

3

```
import csv
blanks = total = count = 0
with open("marks.csv") as f:
    for row in csv.DictReader(f):
        mark = row["maths"]
        if mark == "":
            blanks = blanks + 1
        else:
            total = total + int(mark)
            count = count + 1
print(blanks)
print(total / count)
```

Filled with zeros, the six rows average 43 ÷ 6, about 7.2.

4

In range (0 to 300 mm) — 999, the gauge's error code
Not empty — A reading nobody wrote down
No repeats — Tuesday's reading sent twice
Known name — "Hiltop" instead of Hilltop

Each check catches a different kind of fault, so you need all four.

5

```
import csv
seen = set()
with open("rain.csv") as f:
    for row in csv.DictReader(f):
        key = (row["station"], row["day"])
        if key in seen:
            print(row["station"], row["day"])
        seen.add(key)
```

The key is the station and the day together: one reading per station a day.